

LAMP LLM to ASP Modeling Protocol

Task

ASP rule induction. Given facts F and a target answer set A^* , find a small program Π with

$$AS(F \cup \Pi) = \{A^*\}.$$

Bounded unary normal rules.

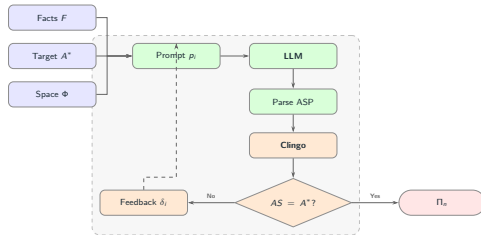
Loop

- LLM proposes rules from F , A^* , schema bounds.
- CLINGO verifies the stable model.
- Mismatches feed back. Up to 3 attempts.
- No fine tuning. Solver decides correctness.

Setup

120 synthetic benchmarks varying predicates, terms, facts, body literals, negation, derived atoms. ILASP baseline runs conflict driven search over an ASP meta encoding, 7 min timeout.

Iterative Loop



Results across 120 benchmarks

System	1st	Final	Fails
<i>gpt-5-mini</i>	90.8%	99.2%	1
<i>gpt-oss</i>	83.3%	98.3%	2
<i>qwen3-coder</i>	19.2%	64.2%	43
ILASP	—	97.5%	3

Output literals 3 to 6 vs 22 or 33 originals. Stratified